View Review

Paper ID
229

Paper Title
Investigating and Constructing an Advanced Nonlinear Deep Learning Model for Crude Oil Price Prediction

REVIEW QUESTIONS

1. Relevance of the Work with the theme of Conference ?
Yes

2. Novelty in the Paper
Yes

3. Comments to authors
The paper presents a compelling approach to crude oil price prediction using a nonlinear deep learning model incorporating LSTMs, GRUs, and transformer-based architectures. The inclusion of macroeconomic indicators and NLP-based sentiment analysis enhances the model's relevance, and the reported performance (MAE: 0.98, RMSE: 1.39) demonstrates superior accuracy compared to traditional models like ARIMA and Random Forest. The methodology is well-structured, with clear
descriptions of data preprocessing, feature engineering, and model evaluation. However, several areas need improvement. The introduction and literature review contain repetitive phrases (e.g., "Beginning with gathering crude oil price data" appears multiple times), which affects readability. The dataset description lacks specifics, such as the time range, frequency (daily/weekly), or size of the dataset, which is critical for reproducibility. The comparison with baseline models (e.g., ARIMA, SVM) is mentioned but lacks detailed quantitative results (e.g., baseline MAE/RMSE values). Additionally, the discussion on model interpretability (e.g., SHAP, LIME) is promising but lacks
implementation details or results, reducing its impact. The repeated reference to "Driver Safety" in the results section appears to be an error and should be removed. I recommend revising the paper to eliminate repetition, provide detailed dataset information, and include quantitative comparisons with baseline models. Elaborating on the interpretability analysis and clarifying the "Driver Safety" error would strengthen the paper. Overall, the work is a strong contribution but requires minor revisions for clarity and rigor.

4. Overall Score- Strongly Accept, Accept, Reject or Borderline
Accept

5. Reviewers Confidence
High

6. Confidential Remarks to the Track Chair
The paper is a strong candidate for acceptance due to its innovative use of hybrid deep